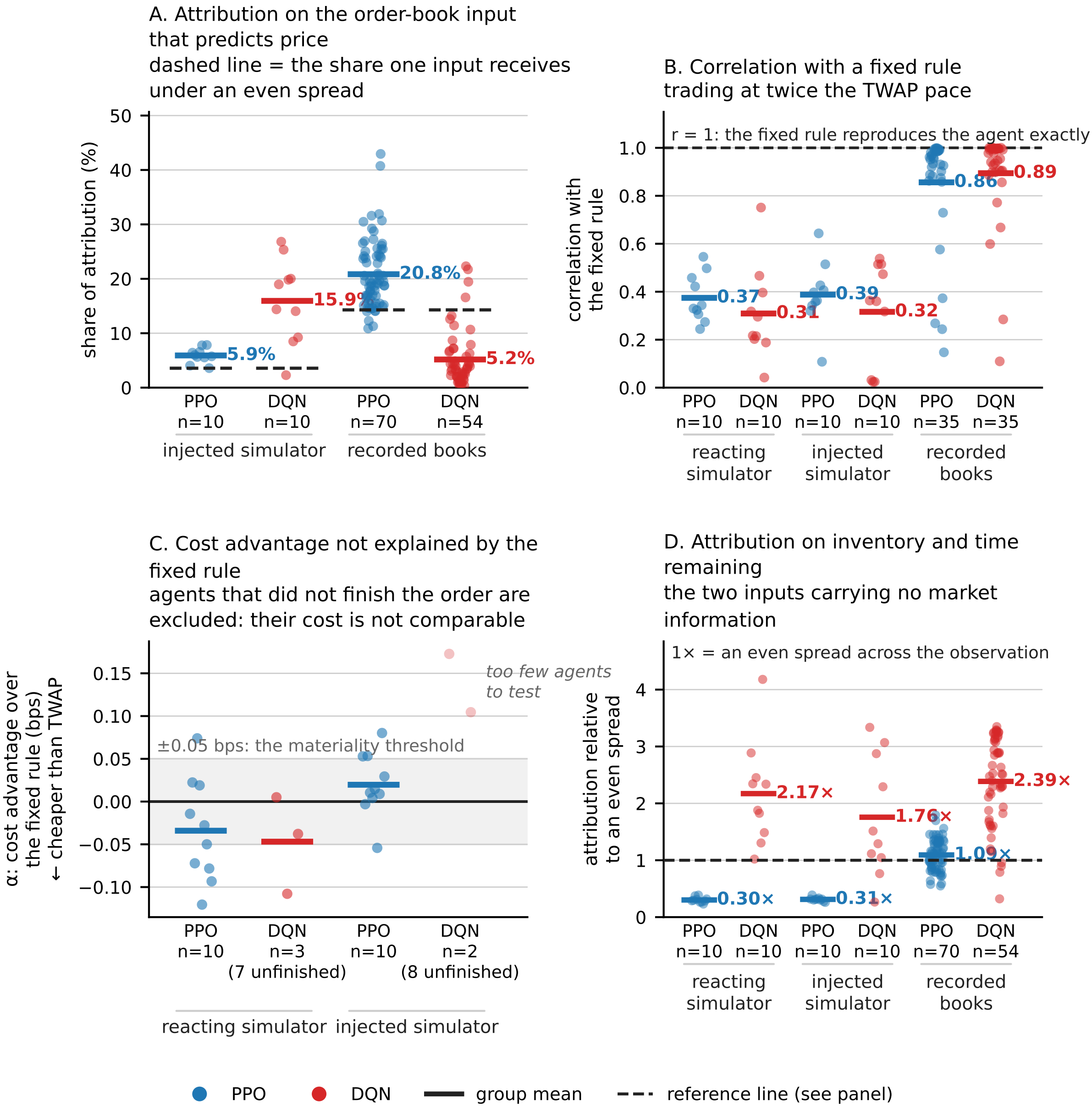

A. Attribution on the order-book input that predicts price
dashed line = the share one input receives under an even spread
share of attribution (%)
5.9%
15.9%
20.8%
5.2%
PPO n=10
DQN n=10
PPO n=70
DQN n=54
injected simulator
recorded books
B. Correlation with a fixed rule trading at twice the TWAP pace
r = 1: the fixed rule reproduces the agent exactly
correlation with the fixed rule
0.37
0.31
0.39
0.32
0.86
0.89
PPO n=10
DQN n=10
PPO n=10
DQN n=10
PPO n=35
DQN n=35
reacting simulator
injected simulator
recorded books
C. Cost advantage not explained by the fixed rule
agents that did not finish the order are excluded: their cost is not comparable
α: cost advantage over the fixed rule (bps) ← cheaper than TWAP
±0.05 bps: the materiality threshold
too few agents to test
PPO n=10
DQN n=3 (7 unfinished)
PPO n=10
DQN n=2 (8 unfinished)
reacting simulator
injected simulator
D. Attribution on inventory and time remaining
the two inputs carrying no market information
1× = an even spread across the observation
attribution relative to an even spread
0.30×
2.17×
0.31×
1.76×
1.09×
2.39×
PPO n=10
DQN n=10
PPO n=10
DQN n=10
PPO n=70
DQN n=54
reacting simulator
injected simulator
recorded books
PPO
DQN
group mean
reference line (see panel)